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Narula Institute of Technology
An Autonomous Institute under MAKAUT
2023
END SEMESTER EXAMINATION - ODD 2023
IT101 - Programming for Problem Solving

TIME ALLOTTED: 3Hours**FULL MARKS: 70**

Instructions to the candidate:

Figures to the right indicate full marks.

Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Group A

(Multiple Choice Type Questions)

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1. File manipulation functions in C are available in which header file ? (1) CO5 BL2
 - a) streams.h
 - b) stdio.h
 - c) stdlib.h
 - d) files.h

2. Which among the following is a unconditional control structure? (1) CO3 BL2
 - a) do-while
 - b) if –else
 - c) goto
 - d) for

3. int **x; means (1) CO3 BL2
 - a) x is a pointer to pointer
 - b) x is not pointer
 - c) x is long
 - d) None of these

4. Which of the following is a keyword used for a storage class? (1) CO1 BL2
 - a) printf
 - b) external
 - c) auto
 - d) scanf

5. The C language consist of _____ number of keywords. (1) CO2 BL2
 - a) 32
 - b) 40
 - c) 24
 - d) 56

6. What is the output of this program ? (1) CO1 BL2

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void main()

{ int a=b=c=10;

a=b=c=50;

printf("\n %d %d %d",a,b,c);

}

```

- a) 50 50 50
- b) Compile Time Error
- c) 10 10 10
- d) Three Garbage Value

7. What is a compiler? (1) CO1 BL1

- a) A compiler does a conversion line by line as the program is run.
- b) A compiler converts the whole of a higher-level program code into machine code in one step.
- c) A compiler is a general-purpose language that provides very efficient execution.
- d) None of the mentions.

8. The advantage of recursion is (1) CO1 BL2

- a) Coding complexity is less.
- b) Time complexity is less.
- c) Space complexity is less.
- d) None of these

9. Which of the following is the first operating system developed using C programming? (1) CO1 BL1

- a) Windows
- b) DOS
- c) Mac
- d) UNIX

10. Which of the following is true for variable names in C? (1) CO1 BL1

- a) Variable names cannot start with a digit
- b) Variable can be of any length
- c) They can contain alphanumeric characters as well as special characters
- d) Reserved Word can be used as variable name

11. The array is the ____ data type in the C programming language. (1) CO1 BL2

- a) Derived Data type
- b) Primitive Data type
- c) Custom Data type
- d) None of these

12. Which statement interrupts the flow of control? (1) CO1 BL4

- a) switch
- b) goto
- c) continue
- d) break

Group B
(Short Answer Type Questions)
(Answer any three of the following) 3x5=15

- | | | | |
|---|-----|-----|-----|
| 13a. What are the differences between structures and arrays? | (4) | CO3 | BL2 |
| 13b. What is 2d array? | (1) | CO1 | BL2 |
| 14. Write a C program to find GCD of two numbers using ternary operator and for loop. | (5) | CO1 | BL2 |
| 15. Write a C code to print all the prime numbers between 1 to 100. | (5) | CO2 | BL3 |
| 16. What is pointer ? Explain how the pointer variable declared and initialized. | (5) | CO4 | BL3 |
| 17. What is meant by variables in C? Explain the rules for defining variables. | (5) | CO1 | BL1 |

Group C
(Long Answer Type Questions)
(Answer any three of the following) 3x15=45

- | | | | |
|---|------|-----|-----|
| 18a. What is structure in C Programming? Differentiate between structure and Union briefly.
Suppose you want to store information about your C programming class of 30: his/her name, roll, and percentage of marks. You can create different variables name, roll and percentage to store this information. Write a C program for displaying this data. | (9) | CO4 | BL3 |
| 18b. What is file in C programming? What is the role of file in C programming? Explain different file handling modes. | (6) | CO5 | BL2 |
| 19a. What is recursion?
A factorial is a function that multiplies a number by every number below it till 1. For example, the factorial of 3 represents the multiplication of numbers 3, 2, 1, i.e. $3! = 3 \times 2 \times 1$ and is equal to 6. Write a recursive C program to find the factorial of 5. | (5) | CO4 | BL6 |
| 19b. Differentiate between "call by value" and "call by address". | (5) | CO4 | BL4 |
| 19c. What is a "function prototype"? Explain "call by value" with a C programming. | (5) | CO4 | BL3 |
| 20a. List the differences between while loop and do-while loop. Write a C program to find sum of Natural numbers from 1 to N using for loop. | (10) | CO3 | BL4 |
| 20b. Difference between array and pointer. | (5) | CO1 | BL2 |
| 21a. What is the syntax of "if else if ladder"?
Grading systems commonly use the letters O, E, A, B, C, D, and F to represent different levels of achievement. In this framework, 'O' typically stands for Outstanding, indicating exceptional performance; 'E' for Excellent, denoting a high level of accomplishment; 'A' for Above Average, reflecting solid competence; 'B' for Average, signifying satisfactory work; 'C' for Below Average, suggesting a need for improvement; 'D' for | (9) | CO3 | BL5 |

Deficient, indicating significant shortcomings; and 'F' for Fail, representing a lack of attainment or inadequate performance. Write a C-program for engineering grading systems by applying "if else if ladder" .

- 21b. Explain "break and continue" statements with suitable C programming. (6) CO3 BL3
22. Write short note: (any three) (15) CO5 BL2
- (i) Garbage values
 - (ii) Header files
 - (iii) Type casting
 - (iv) Universal logic gates
 - (v) Binary file and text file
 - (vi) Bitwise operator